Tutorial Questions

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Question 1

A spherical balloon with a diameter of 6 m is filled with helium at 20°C and 200 kPa. Determine the mole number and the mass of the helium in the balloon. [Answer: 9.28 kmol, 37.2 kg]

Solution

Assumptions Made: We will assume that the gas filling the spherical ballon is an ideal gas, and hence it obeys the ideal gas equation;

$$Pv = RT$$

Note that in this case, R is unique to the helium gas, which we do not know. We can work our way around it by using this equation;

$$PV = nR_uT$$

Note: The v is for specific volume, whiles the V is for actual volume.

In this case, R_u is the universal gas constant which is $8.314kJ/kmolK$. Making n the subject, we get;

$$n = \frac{PV}{R_uT}$$

We have P, Ru and T. Solving for V, we get;

$$V = \frac{4\pi r^3}{3}$$

$$\implies \frac{4\pi 3^3}{3}$$

$$V = 113.0973 \text{ m}^3$$

$$n = \frac{(200)(113.0973)}{(8.314)(20+273)}$$

$$n = 9.277 \text{ kmol} \approx 9.28 \text{ kmol}$$

The mass of the helium in the balloon can be found by multiplying the mole number by the molar mass of helium, which is 4.003kg/kmol

$$m = n \times M$$
$$\implies 9.28 \times 4.003$$
$$m = 37.14 \text{ kg}$$

Question 2

The pressure in an automobile tire depends on the temperature of the air in the tire. When the air temperature is 25°C, the pressure gage reads 210 kPa. If the volume of the tire is 0.025 m3, determine the pressure rise in the tire when the air temperature in the tire rises to 50°C. Also, determine the amount of air that must be bled off to restore pressure to its original value at this temperature. Assume the atmospheric pressure to be 100 kPa.

Solution

Assumption(s) Made: We are going to assume this is an ideal gas, hence it follows the ideal gas equation;

$$Pv = RT$$

The initial pressure of the tire was recorded at 310 kPa (absolute, not gage), with a specific volume of 0.025m^3 and a temperature of 298 K.

When the temperature increases in the tire to 323 K, the volume does not change, it is the pressure that changes. Since R is also equal in the separate conditions, we can make R the subject and equate the two conditions as follows;

$$\frac{P_1 v}{T_1} = \frac{P_2 v}{T_2}$$
$$\implies \frac{P_1}{T_1} = \frac{P_2}{T_2}$$
$$P_2 = \frac{P_1 T_2}{T_1} = 336kPa$$
$$\Delta P = 336 - 310 = 26 \text{ kPa}$$

For the second part, in order to restore it to the initial pressure at the same temperature, some mass of air must be bled off. Before that, the mass of the air when the temperature rose.

$$m_2 = \frac{PV}{RT} = \frac{336 \times 0.025}{0.287 \times 323} = 0.091 \text{ kg}$$

Now, to find the mass of the air when the pressure is restored to 310 kPa;

$$m_3 = \frac{PV}{RT} = \frac{310 \times 0.025}{0.287 \times 323} = 0.084 \text{ kg}$$
$$\Delta m = 0.091 - 0.084 = 0.007 \text{ kg}$$

Question 3

Determine the density, specific gravity, and mass of the air in a room whose dimensions are $4m \times 5m \times 6m$ at $100kPa$ and $25°C$

Solution

Assumptions Made: The air is treated like an ideal gas, hence the ideal gas equation can be used.

$$Pv = RT$$
$$P = \rho RT$$
$$\rho = \frac{P}{RT}$$

Putting in the values, we get;

$$\rho = \frac{100}{(0.287)(25+273)} = 1.169 \text{ kg/m}^3$$

The specific gravity can be calculated by;

$$SG = \frac{\rho}{\rho_{H_2O}} = \frac{1.169}{1000} = 1.169 \times 10^{-3}$$

The mass of the air can be calculated by;

$$m = \rho \times V = 1.169 \times (4 \times 5 \times 6) \text{ kg} = 140.28 \text{ kg}$$

Question 4

Water at 1 atm pressure is compressed to 800 atm pressure isothermally. Determine the increase in the density of water. Take the isothermal compressibility of water to be 4.80×10^{-5} atm^{-1}.

Solution

Using the equation $\frac{\Delta\rho}{\rho} = \alpha\Delta P - \beta\Delta T$; since this is an isothermic compression, the ΔT goes to zero. Therefore;

$$\Delta\rho = \rho \times \alpha\Delta P$$
$$\Delta\rho = 998 \times 4.80 \times 10^{-5} \times (800 - 1)$$
$$\Delta\rho = 38.28 \text{ kgm}^{-3}$$

Question 5

Water at 15˚C and 1 atm pressure is heated to 95˚C at constant pressure. Using coefficient of volume expansion data, determine the change in the density of water. [$\beta = 0.484 \times 10^{-3}$ K^{-1}]

Solution

Using the equation; $\frac{\Delta\rho}{\rho} = \alpha\Delta P - \beta\Delta T$, since this is isobaric heating, ΔT goes to zero. Hence;

$$\Delta\rho = -\rho \times \beta\Delta T$$

$$\Delta\rho = -1000 \times 0.484 \times 10^{-3} \times (95 - 15)$$

$$\Delta\rho = -38.72 \text{ kgm}^{-3}$$

Question 6

The density of seawater at a free surface where the pressure is 98 kPa is approximately 1030 kgm^{-3}. Taking the bulk modulus of elasticity of seawater to be 2.34×10^9 N/m^2 and expressing variation of pressure with depth z as $dP = \rho g dz$ determine the density and pressure at a depth of 2500 m. Disregard the effect of temperature.

Solution

from the question, we see that both the density, as well as the pressure will vary in depth. So we will need to work an equation that will relate the pressure with density, as well as the depth.

Procedure: From the question we know that $dP = \rho g dz$. We also know that $\kappa = \rho(\frac{\partial P}{\partial \rho})_T$. This can also be defined as;

$$\kappa = \rho(\frac{dP}{d\rho})$$

$$\kappa = \rho(\frac{\rho g dz}{d\rho})$$

$$\kappa = \frac{\rho^2 g dz}{d\rho}$$

Grouping like terms give us;

$$\frac{1}{\rho^2} d\rho = \frac{g}{k} dz$$

Integrating this gives us;

$$\int_{\rho_0}^{\rho} \frac{1}{\rho^2} d\rho = \frac{g}{k} \int_0^z dz$$

$$\frac{1}{\rho_0} - \frac{1}{\rho} = \frac{gz}{k}$$

$$\rho = \frac{1}{\frac{1}{\rho_0} - \frac{gz}{k}}$$

Now let us find out how the pressure will also vary with depth.

Using $\kappa = \rho(\frac{dP}{d\rho})$;

$$\frac{1}{\rho}d\rho = \frac{1}{\kappa}dP$$

$$\int_{\rho_0}^{\rho} \frac{1}{\rho}d\rho = \frac{1}{\kappa}\int_{P_0}^{P} dP$$

$$\ln\left(\frac{\rho}{\rho_0}\right) = \frac{1}{\kappa}(P - P_0)$$

$$\ln\left(\frac{\frac{1}{\frac{1}{\rho_0} - \frac{gz}{\kappa}}}{\rho_0}\right) = \frac{1}{\kappa}(P - P_0)$$

$$\kappa \ln\left(\frac{1}{1 - \frac{gz\rho_0}{\kappa}}\right) = P - P_0$$

$$P = P_0 + \kappa \ln\left(\frac{1}{1 - \frac{gz\rho_0}{\kappa}}\right)$$

We have found both our equations relating density and pressure to the depth. Putting in the needed parameters gives us the following answers;

$$\rho = \frac{1}{\frac{1}{1030} - \frac{(9.81)(2500)}{2.34\times10^9 N/m^2}} = 1041.2404 \text{ kg/m}^3$$

$$P = (98 \times 10^3) + (2.34 \times 10^9)\ln\left(\frac{1}{1 - \frac{(9.81)(2500)(1030)}{2.34\times10^9}}\right) = 25.5 \text{ MPa}$$

Question 7

Calculate the mass density, specific weight, specific gravity and specific volume of an oil, 10 m^3 of which weighs 70 kN.

Solution

In finding the density, we will need to simply divide the mass by the volume it occupies, and we can find the mass by dividing its weight by the accelaration due to gravity.

$$W = mg$$

$$m = \frac{W}{g} = \frac{70000}{9.81} = 7135.58 \text{ kg}$$

$$\rho = \frac{m}{V} = \frac{7135.58}{10} = 713.56 \text{ kg/m}^3$$

The specific weight can be found by multiplying the density by the acceleration due to gravity.

$$\gamma = \rho \times g = 713.56 \times 9.81 = 7 \text{ kN/m}^3$$

The specific gravity can be calculated by dividing the density by the density of water; 1000 kg/m^3.

$$SG = \frac{\rho}{\rho_{H_2O}} = \frac{713.56}{1000} = 0.7136$$

$$v = \frac{1}{\rho} = \frac{1}{713.56} = 1.4 \times 10^{-3} \text{ m}^3/\text{kg}$$

Question 8

The velocity distribution of a viscous liquid (dynamic viscosity $\mu = 0.75$ Ns/m^2) flowing over a fixed plate is given by $u = 0.64y - y^2$ (u is velocity in m/s and y is the distance from the plate in m) What are the shear stresses at the plate surface and at $y = 0.34$ m?

Solution

The shear stress is a product of the viscosity and velocity gradient.

$$\tau = \mu \left(\frac{du}{dy} \right)$$

$$\frac{du}{dy} = 0.64 - 2y$$

- At the surface of the plate, y = 0;

$$\tau = 0.75 \times 0.64 = 0.48 \text{ N/m}^2$$

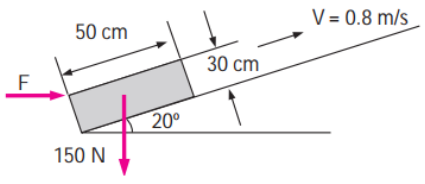
- At y = 0.34 m;

$$\tau = 0.75 \times (0.64 - 2(0.34)) = -0.03 \text{ N/m}^2$$

Question 9

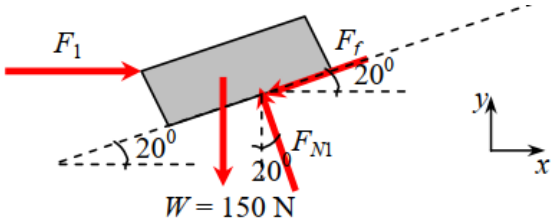
A 50 cm × 30 cm × 20 cm block weighing 150 N is to be moved at a constant velocity of 0.8 m/s on an inclined surface with a friction coefficient of 0.27.

- Determine the force F that needs to be applied in the horizontal direction.
- If a 0.4-mm-thick oil film with a dynamic viscosity of 0.012 Pa · s is applied between the block and inclined surface, determine the percent reduction in the required force.



Solution

For the Figure above, there are four forces in play; the weight, reaction, the friction force and the applied force.



From Newton's first law, summing the forces in the x direction gives us;

$$\Sigma F_x = 0;\ F_1 - F_N \sin(20) - F_f \cos(20) = 0$$

In the y direction, we also get;

$$\Sigma F_y = 0;\ F_N \cos(20) - F_f \sin(20) - W = 0$$

From mechanics, we know that the frictional force is equal to the reaction force times the coefficient of friction.

$$F_f = \mu F_N$$

$$\therefore \Sigma F_x = 0;\ F_1 - F_N \sin(20) - \mu F_N \cos(20) = 0 \quad (1)$$

$$\Sigma F_y = 0;\ F_N \cos(20) - \mu F_N \sin(20) - W = 0 \quad (2)$$

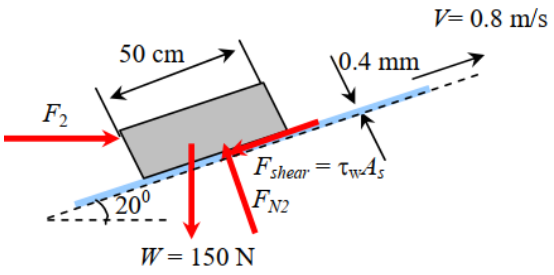
From equation 2, we can make F_N the subject, and this leaves us with;

$$F_N = \frac{W}{\cos(20) - \mu \sin(20)} = 177.023 \text{ N}$$

Putting the value of F_N in equation 1, we get;

$$F_1 = F_N(\sin(20) + \mu \cos(20)) = \mathbf{105.46\ N}$$

In the second instance, oil has been poured on the inclined plane, and hence the frictional force changes, becoming the product of the shear stress and the area of contact.



$$F_f = \tau \times A$$
$$\tau = \mu \left(\frac{V}{h} \right)$$
$$= 0.012 \left(\frac{0.8}{0.4 \times 10^{-3}} \right)$$
$$= 24 \text{ N/m}^2$$
$$F_f = 24 \times (0.5 \times 0.2) = 2.4 \text{ N}$$

Knowing our frictional force now, we can find our reaction force.

$$F_N = \frac{W + F_f \sin(20)}{\cos(20)} = 160.5 \text{ N}$$

The applied force can then be calculated by;

$$F = F_N \sin(20) + F_f \cos(20) = 160.5 \sin(20) + 2.4 \cos(20) = 57.15 \text{ N}$$

The percentage reduction in the force becomes;

$$\text{Percentage reduction} = \frac{105.46 - 57.15}{105.46} \times 100\% = \mathbf{45.8\ \%}$$

Question 10

A 1.9-mm-diameter tube is inserted into an unknown liquid whose density is 960 kg/m^3, and it is observed that the liquid rises 5 mm in the tube, making a contact angle of 15°. Determine the surface tension of the liquid.

Solution

From the capillarity equation;

$$h = \frac{2\sigma_s}{\rho g R} \cos(\phi)$$
$$\sigma_s = \frac{h \rho g R}{2 \cos(\phi)} = \frac{h \rho g d}{4 \cos(\phi)}$$
$$\sigma_s = \frac{(5 \times 10^{-3}) \times 960 \times 9.81 \times (1.9 \times 10^{-3})}{4 cos(15^\circ)} = \mathbf{0.023\ N/m}$$

Question 11

The viscosity of a fluid is to be measured by a viscometer constructed of two 75-cm-long concentric cylinders. The outer diameter of the inner cylinder is 15 cm, and the gap between the two cylinders is 0.12 cm. The inner cylinder is rotated at 200 rpm, and the torque is measured to be 0.8 Nm. Determine the viscosity of the fluid.

Solution

Using the equation;

$$\mu = \frac{60lT}{4\pi^2 R^3 NL}$$

We can put the parameters in and find our viscosity.

$$\mu = \frac{60 \times (0.2 \times 10^{-2}) \times 0.8}{4\pi^2 \times (7.5 \times 10^{-2})^3 \times 200 \times (75 \times 10^{-2})}$$

$$\mu = 0.023 \text{ Pa} \cdot \text{s}$$